IA04: Architecture Decision Record and Failure Walkthrough

Identity: jpkanquandah, team Intellectuals, LiftPro Crane Services Website. Personal repository: https://github.com/Dept-of-Comp-Sci-University-of-Ghana/DE-IND-jpkanquandah. Team repository: https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-intellectuals. Related team issue: (link in the Sakai portfolio). Date: 23 October 2026.

ADR title and status

ADR: Build the website as a static site that opens WhatsApp, with no back end and no database. Status: Accepted.

Context and assumptions

Client problem: Bernard Nyantakyi gets the same questions about services and prices in his DMs and has no single place that shows his services. **Feature:** Enquire button that opens WhatsApp with a prepared message. **Requirement:** a button that sends visitors to his WhatsApp line (D2) and NFR-24 (no customer data stored by the site). **Why the decision matters:** it decides the cost, security, maintenance and how fast we can deliver.

Assumptions and constraints:

- Users: customers on phones, with limited data. Client: not technical, one person.
- Data: services text, images and one WhatsApp number; no customer data.
- Deployment and cost: free or very cheap hosting (the free Vercel plan is limited to non-commercial use, ADR-003).
- Team: 7 students with web skills but a short deadline (a few weeks, four sprints).
- Privacy: avoid collecting personal data. Scale: small, mostly local enquiries.

Alternatives and decision matrix

- **Option A (chosen):** static front end plus a wa.me link.
- **Option B (rejected):** web app with a back end and database that stores enquiries and bookings.
- **Option C (rejected for now):** WhatsApp chatbot and automated replies.

Weights: 5 is very important, 1 is not important. Scores: 1 is poor and 5 is excellent. Weighted score is weight times score.

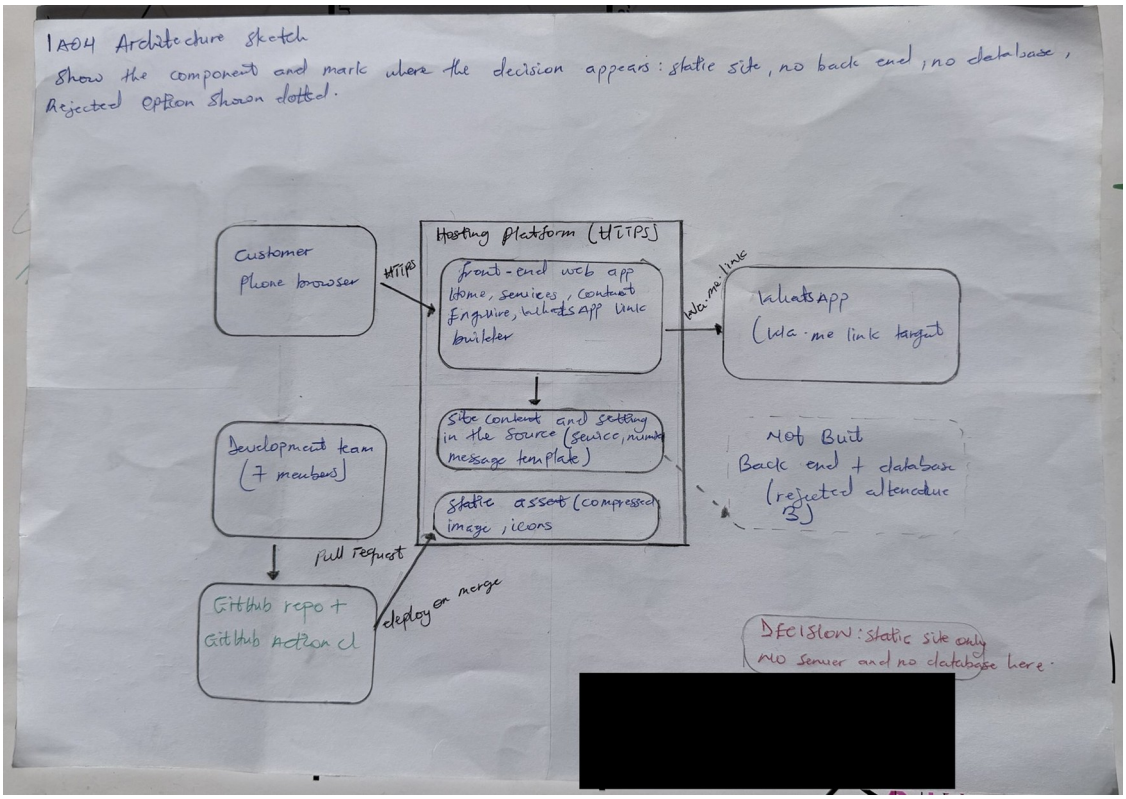
Criterion	Weight	A: static + WhatsApp	B: back end + database	Evidence and reasoning
Fit with client need	5	5 (25)	3 (15)	Client asked only for services plus a WhatsApp button; bookings were not requested

Criterion	Weight	A: static + WhatsApp	B: back end + database	Evidence and reasoning
Security and privacy	5	5 (25)	2 (10)	A stores no customer data; B must protect a database and personal data
Maintainability	4	4 (16)	2 (8)	Fewer parts to break or update for a non-technical client
Team skill and deadline	4	5 (20)	3 (12)	Static pages are quick to finish in four sprints
Deployment and cost	3	4 (12)	2 (6)	Free static hosting versus paying for server and database
Performance and scale	3	5 (15)	3 (9)	Static files are fast and cheap to serve
Total		**113**	**60**	Option C was not scored because it was out of scope (D2)

Decision and balance: Option A. We chose simplicity, privacy and speed over features.

- **Benefits:** very low risk, low cost, no customer data to protect, fast pages.
- **Disadvantages and limits:** no booking, no availability, no prices, and the owner cannot see whether a message was read. Content changes need a maintainer.
- **New work:** a content file kept in source, tests for the link builder, someone trained to update content.
- **Revisit if:** the client wants bookings, prices that change often, or an admin page.

Personal architecture sketch



IA04 handwritten architecture sketch

IA04 handwritten architecture sketch

AI note on this drawing: I drew this by hand myself. The layout was based on a reference sketch that Claude suggested, and I can explain every box and arrow in my own words.

Failure walkthrough

Step	Answer
User action and normal path	Customer taps Enquire; browser follows the link; WhatsApp opens with the message
Failing component	The WhatsApp number in the configuration is wrong after a bad merge
Detection	Tests TC-06 and TC-19 fail in CI; if it slips through, the owner notices enquiries missing or a customer reports it
What the user sees	A chat with the wrong person, or no chat
Data at risk	No stored data; the risk is lost customers and a customer’s message going to a stranger
Logs	CI run logs, the pull request history, the host’s deployment log
Immediate mitigation and recovery	Revert the pull request or redeploy the previous version on the host; confirm by clicking every Enquire link

Step	Answer
Test of the failure handling	A unit test checks every generated link starts with the approved number

Verification plan

1. Run the unit tests for the link builder (make test).
2. Check CI passes on the pull request.
3. Open the deployed site on a phone and click each Enquire button.
4. Ask the client to confirm the number shown and the service names.

Video, commits and declaration

- Video (unlisted, 3 to 4 minutes): https://youtu.be/bMEDxWQgDD4.
- Three meaningful commits made by me as I finished each part: IA4: add jpkanquandah context and alternatives ((link in the Sakai portfolio)); IA4: add jpkanquandah decision and consequences ((link in the Sakai portfolio)); IA4: add jpkanquandah final evidence ((link in the Sakai portfolio)).
- Files: IA04/evidence/architecture-sketch.pdf and IA04/submission/EVIDENCE.md.
- **AI used:** Yes. Claude (Anthropic), 23 October 2026: drafted the decision matrix and wording from our team documents. I checked the scores against our own reasoning, kept the weights, and verified the decision against the client's request in D0 and the hosting choice in our ADR.
- Declaration: "I confirm that this submission is based on my team's actual client project. I have identified all AI assistance used. I can explain the decision, alternatives, trade-offs, failure scenario and evidence in my own words."